

**NANYANG TECHNOLOGICAL UNIVERSITY**  
**SPMS/DIVISION OF MATHEMATICAL SCIENCES**

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 8

For the tutorial in Week 9 on 21 March, let us discuss the following questions.

**Question 1.** Draw and determine whether each of these graphs is strongly connected and if not, whether it is weakly connected.

- (a)  $G = (V, E)$  with  $V = \{a, b, c, d, e\}$ , and directed edges  
 $E = \{(b, a), (b, c), (b, e), (e, a), (d, b), (c, d)\}$ . 
- (b)  $G = (V, E)$  with  $V = \{a, b, c, d, e\}$ , and directed edges  
 $E = \{(a, d), (a, e), (b, a), (b, c), (d, b), (d, c), (e, b), (e, d)\}$ . 
- (c)  $G = (V, E)$  with  $V = \{a, b, c, d, e, f, g\}$ , and directed edges  
 $E = \{(a, c), (a, g), (g, d), (d, c), (e, b), (b, f), (f, e)\}$ . 

**Question 2.** The *degree sequence* of a graph  $G = (V, E)$  is the sequence of length  $n = |V|$  that contains  $\deg(v)$  for all  $v \in V$ . Usually it is sorted non-decreasingly.

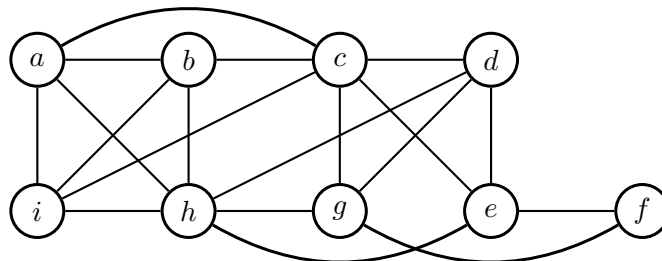
- (a) State the degree sequence of the following graphs:



- (b) Prove that two graphs that do not have the same degree sequence cannot be isomorphic.

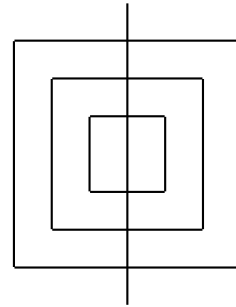
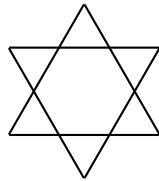
**Question 3.** If a graph  $G$  is connected and has no simple path with length greater than  $k$ , prove that every two simple paths in  $G$  of length  $k$  have at least one vertex in common.

**Question 4. (Ex. 10.5.4.)** Does this graph have an Euler circuit? Find one if there is one.



**Question 5.** Is it true that by adding edges to a simple graph  $G = (V, E)$  (without introducing multiple edges between any pair of vertices) one can always create a graph that has an Euler circuit? Justify the answer.

**Question 6. (Ex. 10.5.7, 8.)** Determine whether each of the picture shown below can be drawn with a pencil in a continuous motion without lifting the pencil or retracing part of the picture.

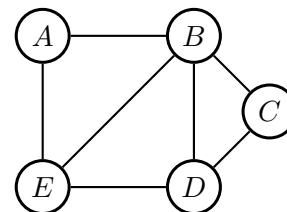
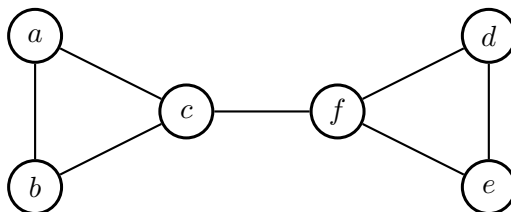


*Profchan's fav*

**Question 7 (Ex. 10.5.16).** For what values of  $n$  do these graphs have an Euler circuit?

- (a)  $K_n$ ,  $\rightarrow$  complete  
(b)  $C_n$ ,  $\rightarrow$  cycle

**Question 8 (Ex. 10.5.20, 21).** Determine whether each of the given graph below has a Hamilton circuit. If it does, find such a circuit. If it does not, give an argument to show why no such circuit exists.




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The following exercise will not be discussed during tutorial. The inequality  $n - k \leq m$  is manageable but the other inequality  $m \leq (n - k)(n - k + 1)/2$  is quite involved. You may ignore this question for test and exam purposes.

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**Question 9.** Given a simple graph  $G$  with  $n$  vertices and  $m$  edges. If  $G$  has exactly  $k$  connected components, prove that

$$n - k \leq m \leq \frac{(n - k)(n - k + 1)}{2}.$$